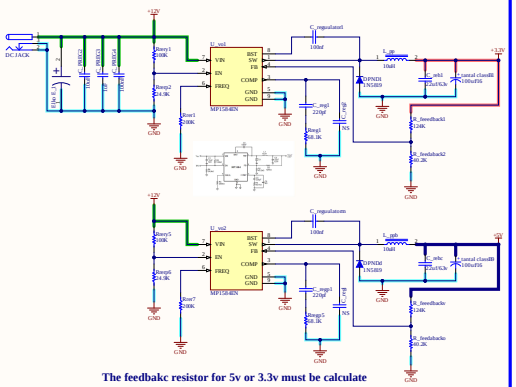
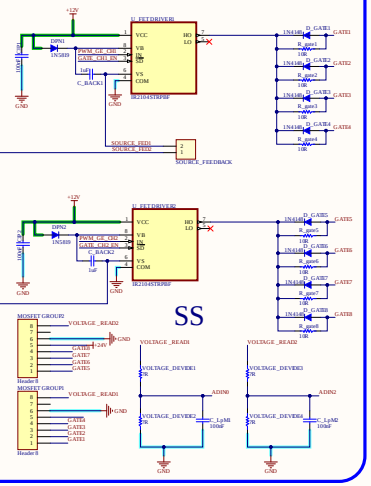
MPS1584 SUPPLY 3.3V/5V



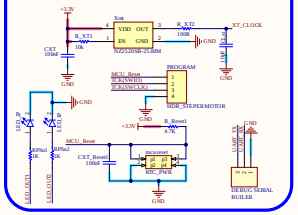
The feedbakc resistor for 5v or 3.3v must be calculate

MOSFET DRIVER IRF2104S

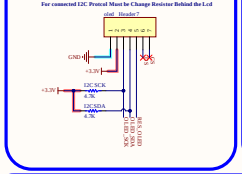


SS

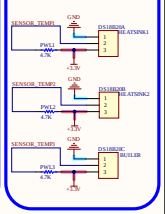
OSCILATOR & PROG



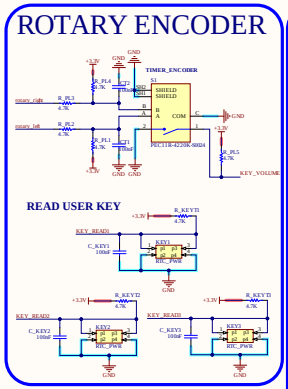
OLED 0.96 INCH



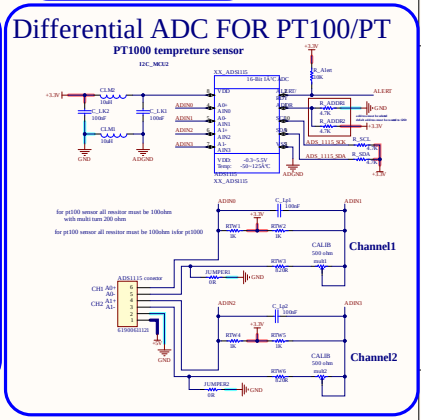
DS18B20



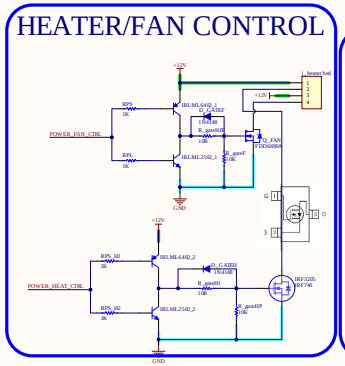
ROTARY ENCODER



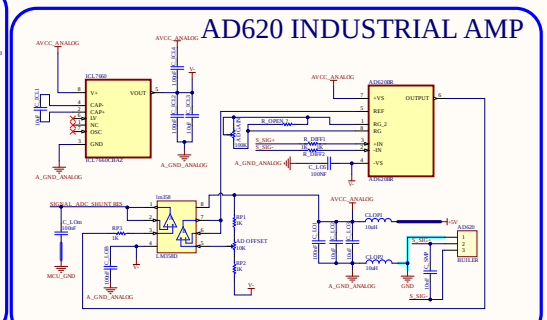
Differential ADC FOR PT100/PT



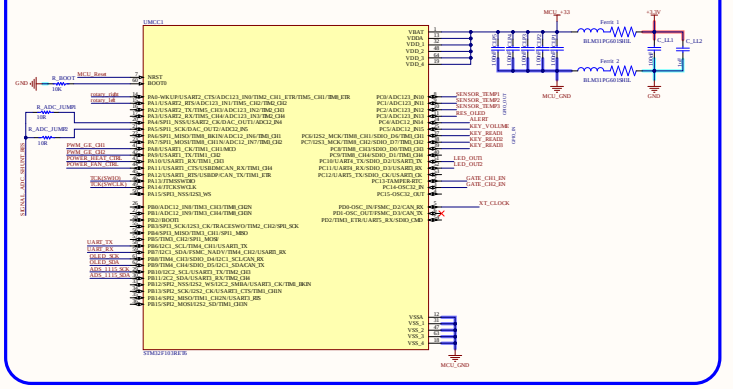
HEATER/FAN CONTROL



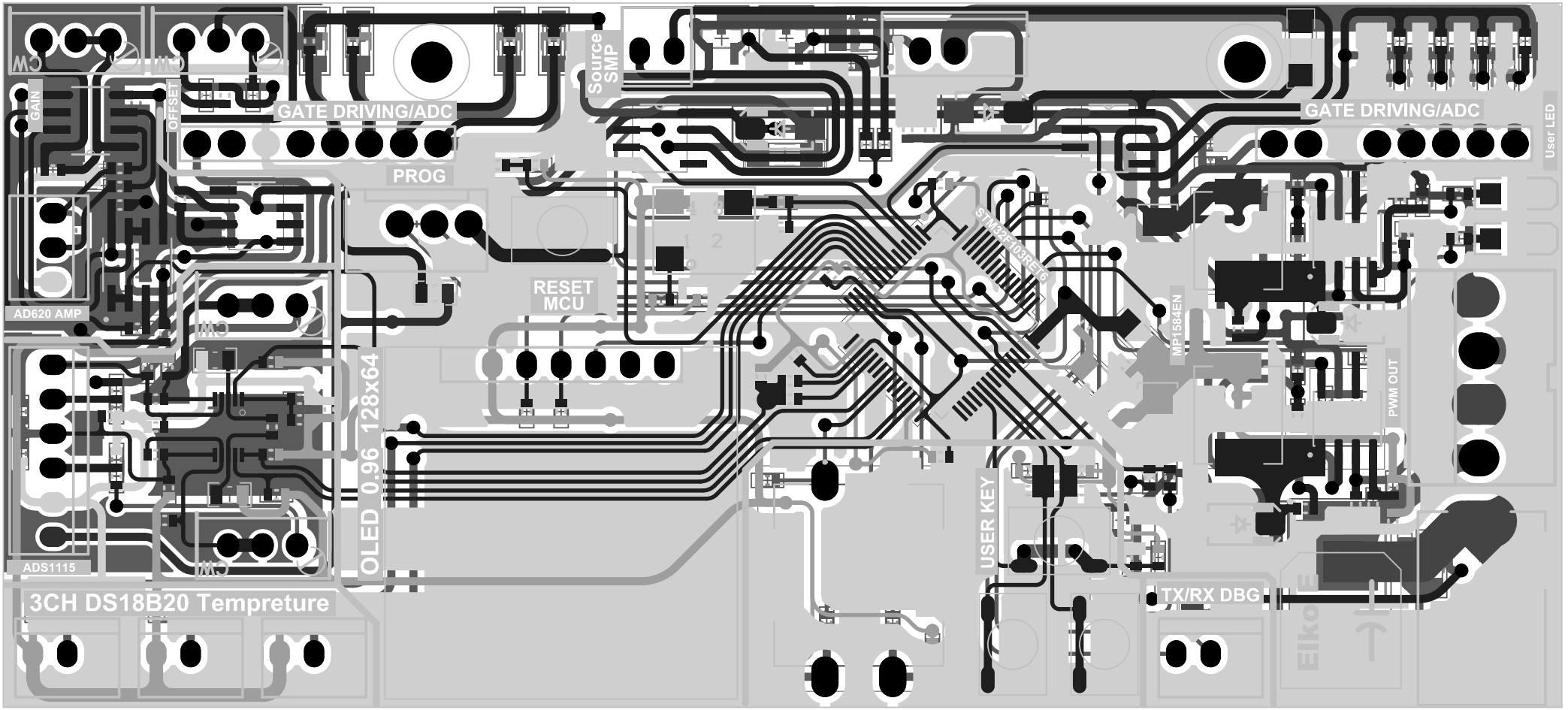
AD620 INDUSTRIAL AMP



MCU STM32F103RET6



Design:Farzan Rezaei


GAIN
OFFSET
GATE DRIVING/ADC
Source SMP
GATE DRIVING/ADC
User LED
PROG
RESET MCU
STM32F103RET6
AD620 AMP
MP1584EN
PWM OUT
OLED 0.96 128x64
USER KEY
ADS1115
3CH DS18B20 Tempreture
TX/RX DBG

Comment	Description	Designator	Footprint	LibRef	Quantity
BULER	WR-WTB 2.54mm Male Locking Header, 3p	AD620, DEBUG SERIAL, DS18B20C	61900311121	61900311121	3
AD620BR	Instrumentation Amplifier 1 Circuit - 8-SOIC	AD620BR	SOIC127P600X175-8N	AD620BR	1
100K	Variable Resistor	AD:GAIN,	Multi Turn	Resistor Variable	1
10K	Variable Resistor	AD:OFFSET	Multi Turn	Resistor Variable	1
61900611121	WR-WTB 2.54mm Male Locking Header, 6p	ADS1115 conector	61900611121	61900611121	1
1uF	CAP 1206	C_BACK1, C_BACK2, C_PREG3	CAPC4817X12N	CAP SMD 1206	3
100nf	Bipolar Capacitor	C_IC1,1, C_IC1,2, C_IC1,3, C_IC1,4, C_KEY1, C_KEY2, C_KEY3, C_LK1, C_LK2, C_LL1, C_LL2, C_LO1, C_LO2, C_LO3, C_LO5, C_LO6, C_LOm, C_Lp1, C_Lp2, C_LpM1, C_LpM2, C_reg1, C_regp1, CIP1, CIP2, CLP1, CLP2, CLP3, CLP4, CLPS, CLxt, CT1, CT2, CXT, CXT_Reset1	SMD_CAP_0603	Cap-Bipolar	35
10uF	CAP 1206	C_PREG2	CAPC4817X12N	CAP SMD 1206	1
100nF	CAP 1206	C_PREG4	CAPC4817X12N	CAP SMD 1206	1
Ns	Bipolar Capacitor	C_regl,C_regl	SMD_CAP_0603	Cap-Bipolar	2
22uf/6.3v	Bipolar Capacitor	C_regulator1, C_regulatorm, C_reh1, C_rehc	SMD_CAP_0603	Cap-Bipolar	4
10nf	Bipolar Capacitor	C_SMP	SMD_CAP_0603	Cap-Bipolar	1
10uH	TOTAL INDUCTOR SMD PACK	CLM1, CLM2, CLOP1, CLOP2	IND0805	INDUCTOR SMD PACKAGE	4
1N4148	General Semiconductor	D_GATE1, D_GATE2, D_GATE3, D_GATE4, D_GATE5, D_GATE6, D_GATE7, D_GATE8, D_GATEF, D_GATEH	Diode 1n4148 SMD	1N4148 SMD	10
DC Power Jack	DC Power Jack. Center Pin 2.5mm. Diameter: 5.7mm	DC JACK	DC Power Jack 1	DC Power Jack	1
1N5819	Schottkey Rectifiers	OPN1, DPN2, DPND1, DPNDd	430X270D	1N5819	4
HEATSINK1	WR-WTB 2.54mm Male Locking Header, 3p	DS18B20A	61900311121	61900311121	1
HEATSINK2	WR-WTB 2.54mm Male Locking Header, 3p	DS18B20B	61900311121	61900311121	1
Elko E package		Elko E_1	Elko E	Elko E package	1
BLM31PG601SH1L	FERRITE BEAD 600 OHM 1206 1LN	Ferrit_1, Ferrit_2	BEADC3216X130N	BLM31PG6013H1L	2
4.7K	Resistor	I2C SCK, I2C SDA, PWL1, PWL2, PWL3, R_ADDR1, R_ADDR2, R_KEYT1, R_KEYT2, R_KEYT3, R_PL1, R_PL2, R_PL3, R_PL4, R_PLS, R_Reset1, R_SCL, R_SDA	RES 0603 (1608X09L)	Resistor	18
ICL7660CBAZ	Charge Pump Switching Regulator IC Positive or Negative Fixed -Vin, 2Vin 1 Output 45mA 8-SOIC (0.154, 3.90mm Width)	ICL7660	SOIC127P600X175-8N	ICL7660CBAZ	1
IRF740	MOSFET N-CH 400V 10A TO-220AB	IRF3205	TO-220 AB	IRF740	1
BC337	45 V, 500 mA NPN general-purpose transistors	IRLML2502_1, IRLML2502_2	SOT23_M	BC337	2
BC856	PNP General-purpose Transistor	IRLML6402_1, IRLML6402_2	SOT23_M	BC856	2
	WR-WTB 3.96mm Male Locking Header, 4p	l_heater bed	645004114822	645004114822	1
0R	Resistor	JUMPER1, JUMPER2	RES 0603 (1608X09L)	Resistor	2
RTC_PWR		KEY1, KEY2, KEY3, mcu reset	push_bt	push_button	4
10uH	4.7 ÂµH Shielded Drum Core, Wirewound Inductor 6 A 33.5mOhm Max Nonstandard	l_pp, L_ppb	IND_SRP6540-4R7M	SRP6540-4R7M	2
	LED	LED_1P, LED_2P	LED SMD 1206	LED	2
LM358D	Op Amp Dual Low Power Amplifier Â±15V/30V 8-Pin SOIN Tube	lm358	SOIC127P599X175-8N	LM358D	1
Header 8	Header, 8-Pin	MOSFET GROUP1, MOSFET GROUP2	HDR1X8	Header 8	2
500 ohm	Variable Resistor	multi1, multi2	Multi Turn	Resistor Variable	2
Header 7	Header, 7-Pin	oled	HDR1X7	Header 7	1
HDR_STEPER MOTOR	Header, 4-Pin Telecom.	PROGRAM	CON WF_4	Header 4 PIN Telecom.	1
FDD6690A	N-channel MOSFET	Q_FAN	TO252-3L	FDD6690A	1
10R	Resistor	R_ADC_JUMP1, R_ADC_JUMP2, R_gate1, R_gate2, R_gate3, R_gate4, R_gate5, R_gate6, R_gate7, R_gate8, R_gateHG, R_gateHSF	RES 0603 (1608X09L)	Resistor	12
10K	Resistor	R_Alert, R_BOOT, R_gateF, R_gateHP, R_XT1	RES 0603 (1608X09L)	Resistor	5
1K	Resistor	R_DIFF1, R_DIFF2, RP1, RP2, RP3, RPL, RPS, RPS_H1, RPS_H2, RPSn1, RPSn2, RTW1, RTW2, RTW4, RTW5	RES 0603 (1608X09L)	Resistor	15
40.2K	Resistor	R_feedaback2, R_feedabacko	RES 0603 (1608X09L)	Resistor	2
124K	Resistor	R_feedback1, R_feedbackv	RES 0603 (1608X09L)	Resistor	2
?	Resistor	R_OPEN	RES 0603 (1608X09L)	Resistor	1
100R	Resistor	R_XT2	RES 0603 (1608X09L)	Resistor	1
68.1K	Resistor	Rreg1, Rreg5	RES 0603 (1608X09L)	Resistor	2
200K	Resistor	Rrer1, Rrer7	RES 0603 (1608X09L)	Resistor	2
24.9K	Resistor	Rreg2, Rreg6	RES 0603 (1608X09L)	Resistor	2
100K	Resistor	Rrery1, Rrery5	RES 0603 (1608X09L)	Resistor	2
820R	Resistor	RTW3, RTW6	RES 0603 (1608X09L)	Resistor	2
PEC11R-4220K-S0024	24 Pulse Incremental Mechanical Rotary Encoder w/ 6 mm Knurl Shaft, Through Hole | Bourns PEC11R-4220K-S0024	S1	KDCR_PEC11R-4220K-S0024	PEC11R-4220K-S0024	1
61900211121	WR-WTB 2.54mm Male Locking Header, 2p	SOURCE_FEEDBACK	61900211121	61900211121	1
100uf/16	Electrolytic Capacitor	tantal class81, tantal class89	TANTAL-CAP-Size B	Capacitor Electrolytic	2
IR2104STRPBF	Half-Bridge Gate Driver IC Non-Inverting 8-SOIC	U_FET DRIVER1, U_FET DRIVER2	SOIC127P600X175-8N	IR2104STRPBF	2
MP1584EN	Buck Switching Regulator IC Positive Adjustable 0.8V 1 Output 3A 8-SOIC (0.154, 3.90mm Width) Exposed Pad	U_vo1, U_vo2	SOIC127P600X170-9N	MP1584EN	2
STM32F103RET6	ARM® Cortex®-M3 STM32F1 Microcontroller IC 32-Bit 72MHz 512KB (512K x 8) FLASH 64-LQFP (10x10)	UMCC1	QFP50P1200X1200X160-64N	STM32F103RET6	1
7R	Resistor	VOLTAGE_DEVIDE1, VOLTAGE_DEVIDE2, VOLTAGE_DEVIDE3, VOLTAGE_DEVIDE4	RES 0603 (1608X09L)	Resistor	4
NZ2520SB-25.00M	Oscillator 25.00MHz ±50PPM, 3.3V, Temp -55 to +125°C	Xmt	VCO	OSC_25M_NZ2520SB	1
XX_ADS1115		XX_ADS1115	MSOP10	ADS1115	1